Ex.No:8 NMAP to Discover live hosts using Nmap scans on the TryHackMe Platform.

Aim:

To discover live hosts using Nmap scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform Room.

Introduction:

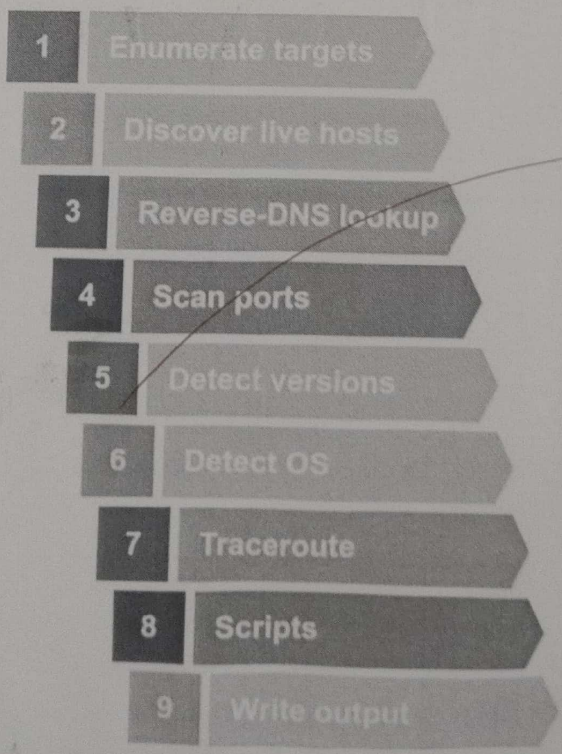
This room is the first of four in this Nmap series. These four rooms are also part of the Network Security module.

⇒ Nmap live host discovery.
⇒ Nmap basic port scans.
⇒ Nmap advanced Port scans.
⇒ NMAP post port scans.

ARP scan: This scan uses ARP requests to discover live hosts.

ICMP scan: This scan uses ICMP requests to identify live hosts.

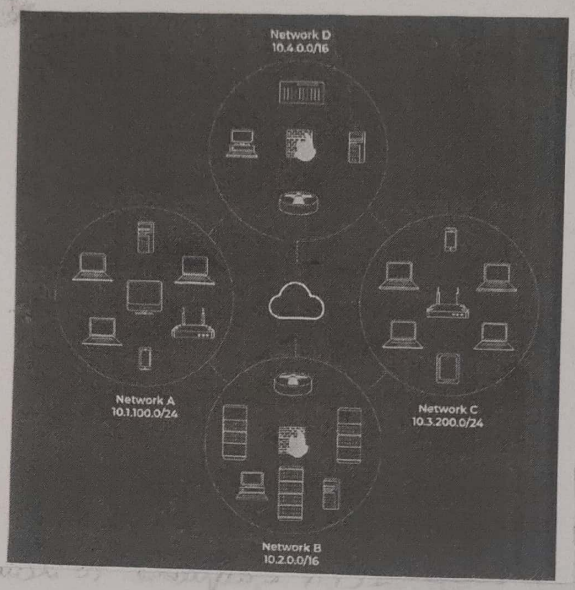
TCP/UDP ping scan: This scan sends packets to TCP ports and UDP ports to determine live hosts.

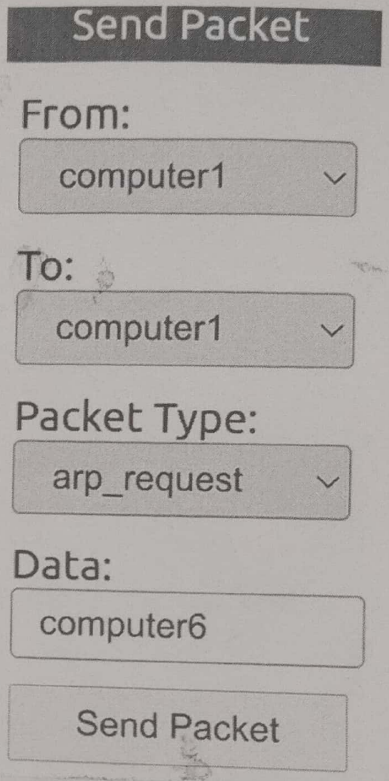


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Subnetworks:

A network segment is a group of computers, linked through a shared medium, like an ethernet switch or WiFi access point. In IP networks, a subnetwork typically consists of one or more network segments connected together and configured to use the same router. A network segment is a physical connection, while a subnetwork is a logical connection.



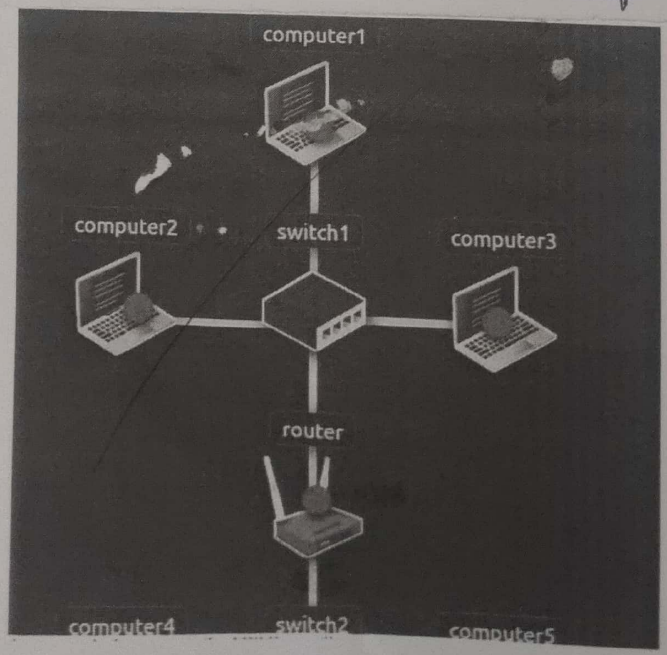


i) How many devices can see the ARP request?

⇒ 4.

ii) Did computer 6 receive the ARP request? (Y/N).

⇒ N.



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i) How many devices can see the ARP request?

⇒ 4

ii) Did computer 6 reply to the ARP request? (Y/N)

⇒ Y.

Discovering Live hosts:

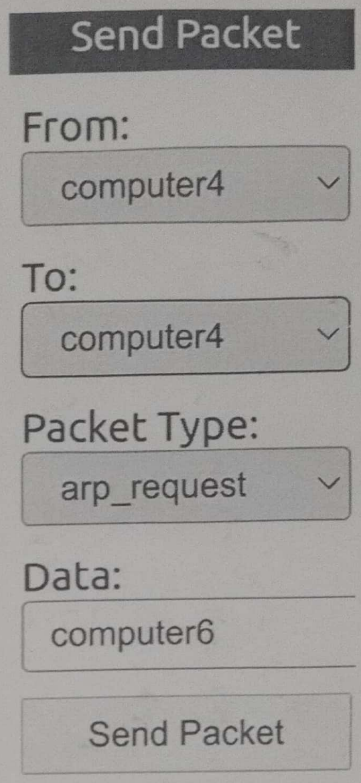
TCP/IP layers:

⇒ ARP from link layer,

⇒ ICMP from Network layer.

⇒ TCP from Transport layer.

⇒ UDP from Transport layer



What is the name of the first device that responded to the first ARP request?

⇒ router,

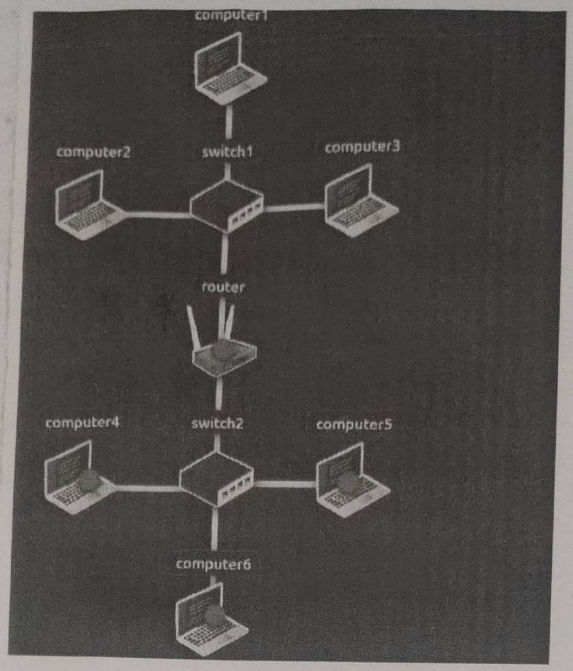
What is the name of the first device that responded to the second ARP request?

⇒ Computer 5,

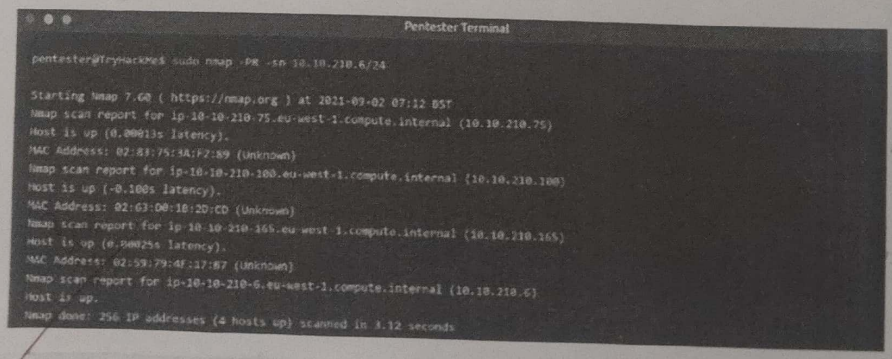
Send another ping request. Did it require new ARP requests? (Y/N).

⇒ N.

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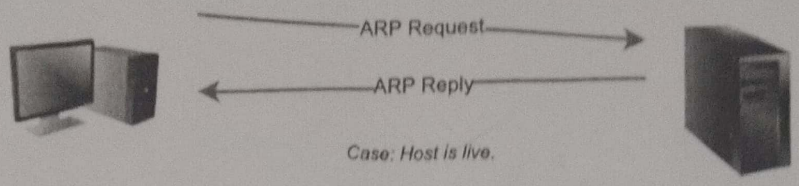


7	Application Layer			
6	Presentation Layer	Application Layer	HTTP, HTTPS, SMTP, POP3, IMAP, SSH, FTP, SNMP, Telnet, RDP,...	
5	Session Layer			
4	Transport Layer	Transport Layer	TCP, UDP	
3	Network Layer	Network Layer	IPv4, IPv6, ICMP, IPsec	
2	Data Link Layer	Link Layer	ARP, Ethernet (802.3), WiFi (802.11), DSL, Bluetooth,	
1	Physical Layer			



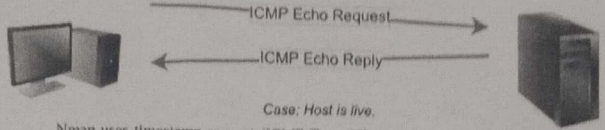
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nmap -PR -sn TARGET



Case: Host is live.

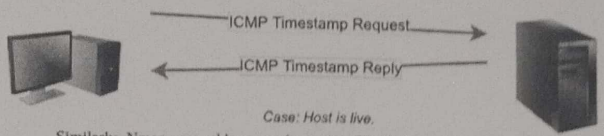
nmap -PE -sn TARGET



Case: Host is live.

Nmap uses timestamp request (ICMP Type 13) and checks whether it will get a Timestamp reply (ICMP Type 14). Adding the -PP option tells Nmap to use ICMP timestamp requests. As shown in the figure below, you expect live hosts to reply.

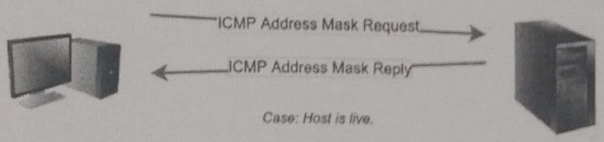
nmap -PP -sn TARGET



Case: Host is live.

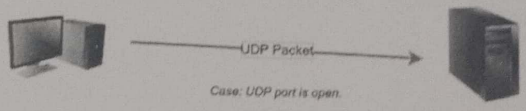
Similarly, Nmap uses address mask queries (ICMP Type 17) and checks whether it gets an address mask reply (ICMP Type 18). This scan can be enabled with the option -PM. As shown in the figure below, live hosts are expected to reply to ICMP address mask requests.

nmap -PM -sn TARGET



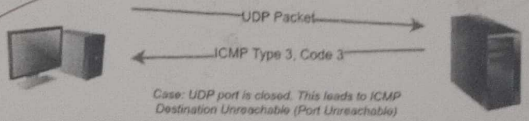
Case: Host is live.

nmap -PU -sn TARGET



Case: UDP port is open.

nmap -PU -sn TARGET



Case: UDP port is closed. This leads to ICMP Destination Unreachable (Port Unreachable)

Answer the questions below

Which TCP ping scan does not require a privileged account?

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